Incident Response and Digital Forensics Lab:
Data hiding in *slack space* and forensic analysis

Mohamed Amin AYED

I.

1. Used *dd* to copy 10 bytes at a rate of 1M at a time from /dev/null (which streams nullbytes) to output file: virtual_disk.img

```
┌──(amin㉿41Gb)-[~]
└─$ dd if=/dev/zero of=virtual_disk.img bs=1M count=10 status=progress
10+0 records in
10+0 records out
10485760 bytes (10 MB, 10 MiB) copied, 0.00560912 s, 1.9 GB/s
```

2. Used *lsop* to simulate physically attaching virtual_disk.img to the hardware -> bytes readable

```
┌──(amin㉿41Gb)-[~]
└─$ sudo losetup /dev/loop10 virtual_disk.img
[sudo] password for amin:

┌──(amin㉿41Gb)-[~]
└─$ ls /dev/lo
log           loop1         loop2         loop4         loop6         loop-control
loop0         loop10        loop3         loop5         loop7
┌──(amin㉿41Gb)-[~]
└─$ ls /dev/loop10
/dev/loop10

┌──(amin㉿41Gb)-[~]
└─$ ls /dev/loop10 -la
brw-rw---- 1 root disk 7, 10 Nov  9 15:22 /dev/loop10
```

:block

Formatting it with EXT4 with 4096-byte block size

```
┌──(amin㉿41Gb)-[~]
└─$ sudo mkfs.ext4 -b 4096 /dev/loop10
mke2fs 1.47.4 (6-Mar-2025)
Discarding device blocks: done
Creating filesystem with 2560 4k blocks and 2560 inodes

Allocating group tables: done
Writing inode tables: done
Creating journal (1024 blocks): done
Writing superblocks and filesystem accounting information: done
```

Verification:

```
┌──(amin㉿41Gb)-[~]
└─$ sudo dumpe2fs /dev/loop10 | grep "Block size"
dumpe2fs 1.47.4 (6-Mar-2025)
Block size:               4096
```

Now time to mount it:

Mounting is making files in the block device readable by the filesystem:

```
┌──(amin㉿41Gb)-[~]
└─$ sudo mkdir -p /mnt/virtual_disk

┌──(amin㉿41Gb)-[~]
└─$ sudo mount /dev/loop10 /mnt/virtual_disk
```

Let's copy this image in

```
┌──(amin㉿41Gb)-[~]
└─$ timg -p 's' /usr/share/pixmaps/debian-logo.png
1;22;23;24;28;32;42;52c

┌──(amin㉿41Gb)-[~]
└─$ sudo cp /usr/share/pixmaps/debian-logo.png /mnt/virtual_disk/

┌──(amin㉿41Gb)-[~]
└─$ ls /mnt/virtual_disk/
debian-logo.png  lost+found
```

the file size is 1206 bytes:

```
┌──(amin㉿41Gb)-[~]
└─$ ls -l /mnt/virtual_disk/debian-logo.png
-rw-r--r-- 1 root root 1206 Nov  9 15:35 /mnt/virtual_disk/debian-logo.png
```

To calculate slack space, it s the size left from the last block used by the file

slackspace = block size - file_size mod(blocksize = 4096 – 1206 = 2890

II.

```
┌──(amin㉿41Gb)-[~]
└─$ sudo umount /mnt/virtual_disk

┌──(amin㉿41Gb)-[~]
└─$ sudo debugfs /dev/loop10
debugfs 1.47.4 (6-Mar-2025)
debugfs:  stat /debian-logo.png
```

```
Inode: 13   Type: regular    Mode:  0644   Flags: 0x80000
Generation: 367782521    Version: 0x00000000:00000002
User:     0   Group:     0   Project:     0   Size: 1206
File ACL: 0
Links: 1   Blockcount: 8
Fragment:  Address: 0    Number: 0    Size: 0
 ctime: 0x6af1da15:0f9eb9dc -- Mon Nov  9 15:35:01 2026
 atime: 0x6af1da15:0e604268 -- Mon Nov  9 15:35:01 2026
 mtime: 0x6af1da15:0f9eb9dc -- Mon Nov  9 15:35:01 2026
crtime: 0x6af1da15:0e604268 -- Mon Nov  9 15:35:01 2026
Size of extra inode fields: 32
Inode checksum: 0x92b8e2f2
EXTENTS:
(0):2048
(END)
```

There is 1 extent; an **extent** is a contiguous range of physical blocks allocated to a file

Block count = 8*512/4096=1 blocks

0th block of the file maps to the 2048th physical block due to other “metadata” holding the first blocks:

- ⇨ So the offset is 2048*4096+size mod(4096) = 8,388,608 bytes + 1206 bytes = 8,389,814 bytes

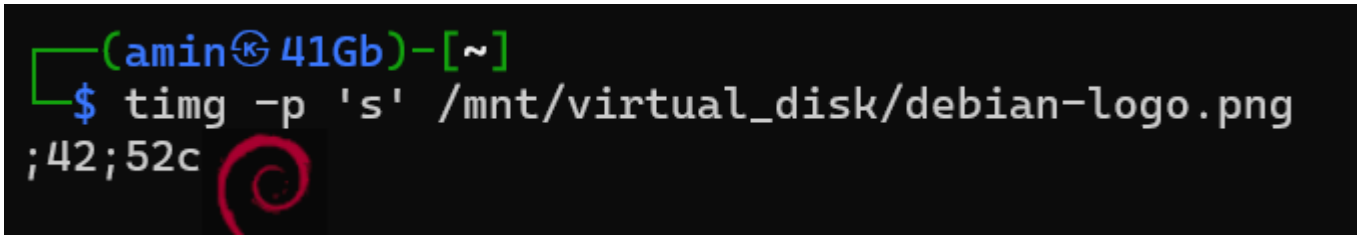
```
echo "this is a hidden message" | sudo dd of=/dev/loop10 bs=1 seek=8389814 conv=notrunc
```

we set offset as 8389814 for bs=1

Let’s write a hidden text at that offset

```
┌──(amin㉿41Gb)-[~]
└─$ echo "this is a hidden message" | sudo dd of=/dev/loop10 bs=1 seek=8389814 conv=notrunc
[sudo] password for amin:
Sorry, try again.
[sudo] password for amin:
25+0 records in
25+0 records out
25 bytes copied, 0.000384934 s, 64.9 kB/s
```

And try to display the image after mounting the virtual disk, and it s intact!



And strings doesn’t show the text we inserted right after it:

```
┌──(amin㉿41Gb)-[~]
└─$ strings /mnt/virtual_disk/debian-logo.png
IHDR
}IDATx
PF)P
Q!+(
!{UN[nU
->{[
9K3*
 gs#
ixXa
P:LH
mcfe
7!!8YT
\Lo&
BmDu
IEND
```

And that is due to the fact that our text is placed in the slack space which is ignored by the EXT4 Filesystem

However using a hex editor on the unmounted disk as a file, we can see the hidden message immediately after END of the file:

```
┌──(amin㉿41Gb)-[~]
└─$ sudo umount /mnt/virtual_disk

┌──(amin㉿41Gb)-[~]
└─$ sudo hexdump -C -s 8389800 -n 64 /dev/loop10
008004a8  c2 ff 00 00 00 00 49 45  4e 44 ae 42 60 82 74 68  |......IEND.B`.th|
008004b8  69 73 20 69 73 20 61 20  68 69 64 64 65 6e 20 6d  |is is a hidden m|
008004c8  65 73 73 61 67 65 0a 00  00 00 00 00 00 00 00 00  |essage..........|
008004d8  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  |................|
008004e8
```

Last block:

```
┌──(amin㉿41Gb)-[~]
└─$ sudo dd if=/dev/loop10 bs=4096 skip=2048 count=1 | hexdump -C
00000000  89 50 4e 47 0d 0a 1a 0a  00 00 00 0d 49 48 44 52  |.PNG........IHDR|
00000010  00 00 00 30 00 00 00 30  08 06 00 00 00 57 02 f9  |...0...0.....W..|
00000020  87 00 00 04 7d 49 44 41  54 78 da ed 59 03 b4 2c  |....}IDATx..Y..,|
00000030  47 10 bd b1 8d 7f 62 3b  99 ae de e7 d8 b6 6d db  |G.....b;......m.|
00000340  c3 98 47 85 ee 3d 8d 7f  94 d1 38 71 42 ab ad 4a  |..G..=....8qB..J|
00000350  bf 76 90 6d e8 4a 0a bb  3e 12 80 49 8f 6e e7 2b  |.v.m.J..>..I.n.+|
00000360  be 07 a9 c3 90 17 5c 4c  6f 26 b6 93 24 d5 40 21  |......\Lo&..$.@!|
00000370  e6 9a 3f 82 e9 f4 7a e6  77 0a b7 56 2e e3 99 eb  |..?...z.w..V....|
00000380  71 3f d7 f5 c9 7d 3e 81  1a 8d 69 a1 35 d3 f0 3a  |q?...}>...i.5..:|
00000390  ab 35 df 5e dc 90 92 38  84 7c 44 4d 13 c2 2e 4a  |.5.^...8.|DM...J|
000003a0  59 e2 69 f2 68 85 dc eb  e7 f8 92 ad 9a 3e ef 09  |Y.i.h........>..|
000003b0  29 ec 7b 59 02 99 20 fd  85 90 c3 d8 1f 52 98 73  |).{Y.. ......R.s|
000003c0  15 f2 52 4d cd fb 03 8b  25 07 bb 7d e2 3a 23 bb  |..RM....%..}.:#.|
000003d0  9c 30 3f d3 af e9 22 e9  2d 19 cc 42 6d 44 75 cb  |.0?...".-..BmDu.|
000003e0  ce 06 59 28 82 88 17 7b  fd 0b 26 a4 9a 02 fb ac  |..Y(...{..&.....|
000003f0  bc ed c8 cb 30 d7 f4 3e  39 f5 3b 7c e3 49 5e 8d  |....0..>9.;|.I^.|
00000400  09 ab 7b 7c 90 15 07 21  64 b3 98 05 3e 21 4d a2  |..{|...!d...>!M.|
00000410  40 f0 ad 0e f3 72 6c 13  9b a3 68 e0 4d 4b 6c 03  |@....rl...h.MKl.|
00000420  4f f8 00 2c 16 14 72 59  b5 dd e7 d5 66 b1 40 06  |O..,..rY....f.@.|
00000430  a9 e6 05 16 2d 9e bb 8b  05 f2 77 ec 72 e2 41 df  |....-.....w.r.A.|
00000440  a5 2b 16 fc 65 de 9f 15  5a 35 3b a1 88 f0 5d b6  |.+..e...Z5;...].|
00000450  5f 99 a1 59 f8 30 43 17  95 99 be f1 62 ef 16 87  |_..Y.0C.....b...|
00000460  39 a7 2a a0 3b 05 73 56  73 04 0b 20 2f 9b 8b 05  |9.*.;.sVs.. /...|
00000470  7f 61 77 a4 bf e7 7a 5b  61 8e 2d 9c dc f0 aa 73  |.aw...z[a.-....s|
00000480  3a 76 a1 15 f2 a3 c2 de  e7 3b ca c5 c3 a5 e8 1a  |:v.......;......|
00000490  97 57 52 ac 01 42 c8 f1  15 2d 15 94 9a 74 eb 86  |.WR..B...-...t..|
000004a0  e3 1f 4c 5f f4 17 58 d9  c2 ff 00 00 00 00 49 45  |..L_..X.......IE|
000004b0  4e 44 ae 42 60 82 74 68  69 73 20 69 73 20 61 20  |ND.B`.this is a |
000004c0  68 69 64 64 65 6e 20 6d  65 73 73 61 67 65 0a 00  |hidden message..|
000004d0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  |................|
*
```

Analyzing the fs with icat yields only the image:



and not the secret message!

EOF